

CapFinLoan - High Level Design (HLD)

1. Executive Summary

The **CapFinLoan** platform is a modern fintech solution designed to manage the end-to-end lifecycle of loan applications. It provides isolated pathways for secure applicant registration, KYC profile setup, remote document verification, and a comprehensive backend queue system for administrators to review and finalize loan decisions. It is built for scale, resilience, and maintainability.

2. System Architecture Paradigm

The system is constructed using a **Distributed Microservices Architecture**. To handle independent scalability and enforce strict logical isolation, business capabilities are sliced into four fundamental backend components connected via an asynchronous event bus and shielded by an edge API Gateway.

3. Technology Stack

- **Backend Framework:** .NET 10 / C# (ASP.NET Core Web API)
- **Frontend Application:** React / TypeScript (Vite)
- **Gateway:** Ocelot API Gateway
- **Database System:** Microsoft SQL Server 2022 (EF Core Code-First mapped)
- **Message Broker:** RabbitMQ
- **Containerization:** Docker & Docker Compose
- **Security & Auth:** Stateless JWT Authentication with BCrypt password hashing.

4. Core Microservices Overview

4.1. Ocelot API Gateway (Edge Routing)

- Acts as the single point of entry for all external client traffic (web and mobile apps).

- Secures the internal network by resolving route aliases (e.g., `/gateway/applications/*` mapping dynamically to internal IP/port setups) and enforcing JWT inspection at the network edge.

4.2. Identity & Auth Service

- Manages user onboarding, Google OAuth integrations, and Email-based OTP flows.
- Acts as the custodian of the User KYC profile (banking details, physical address, and basic identity attributes).
- Independent Database: `CapFinLoanAuth`

4.3. Loan Application Service

- Orchestrates the active creation, tracking, and modification of loan applications by the borrower.
- Manages the "Application Submission Saga"—a distributed orchestration engine ensuring an application passes successfully to the Admin pool without system failure.
- Independent Database: `CapFinLoanApplication`

4.4. Document Service

- Provides secure asset storage and retrieval. Binds applicant-uploaded PDF/JPG/PNG artifacts securely to application scopes.
- Broadcasts storage notifications to the event bus.
- Independent Database: `CapFinLoanDocument`

4.5. Admin Processing Service

- The primary backend driver for loan administrators. Provides high-level visibility over the ecosystem.
- Generates analytic dashboards, manages active queues, and finalizes `Under Review`, `Approved`, or `Rejected` decisions. Automatically synchronizes decisions upstream via RabbitMQ.
- Independent Database: `CapFinLoanAdmin`

5. Information Flow & Messaging

To eliminate horizontal coupling between the Microservices, an **Event-Driven Architecture (EDA)** is heavily employed utilizing **RabbitMQ**.

1. When a user submits an application, `ApplicationService` drops an `application.submitted` event into the message broker.
2. When a borrower uploads a document, `DocumentService` drops a `document.uploaded` event.
3. The `AdminService` subscribes to these streams without holding real-time dependencies on the originating services. It buffers them asynchronously to construct its internal queue.
4. Once an Admin issues an approval, a reciprocal `admin.decision.made` event propagates backward down the message queue to finalize the status dynamically.

6. Security Boundaries

- **Database-per-service:** Microservices do not share database schemas. They cross document updates solely via safe, documented events.
- **Role-Based Authorization Policies:** Routes natively evaluate user identity claims preventing horizontal escalation (Applicant vs Administrator access rights are strictly delineated).